IE305 SIMULATION LAB · FINAL PRESENTATION

RakeFlow

Simulating the Ghugus–Tadali–Chandrapur closed-circuit coal rake cycle in AnyLogic

4.77 → 7.76

rakes/day to the power station

40.3 → 24.7 h

mean rake cycle time

₹33,300 → ₹0

demurrage per rake

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EXECUTIVE SUMMARY

Three terminal changes raise throughput 63% without new track

1. **Time is lost at the terminals, not on the line.**
   80% of the 40.3 h cycle is spent at Ghugus and Chandrapur; the single line and Tadali merge cost 0.6 h.

2. **Ordering the crew during loading adds 1.0 rake/day at no cost.**
   Loaded rakes stop holding one of only two Ghugus loops while waiting for a crew.

3. **A third loop adds another 1.0; a loading silo ends demurrage.**
   Together the three reach 7.76 rakes/day, matching the 7–8 reported for Ghugus.

4. **The ₹228 crore doubling adds only 0.11 rakes/day for today's traffic.**
   The single line is 21% busy; its value is for future growth, not this circuit.

+63%

rakes/day (4.77 → 7.76)

crew timing + 3rd loop + loading silo

−39%

mean cycle (40.3 → 24.7 h)

P90 47.1 → 28.6 h

+2%

doubling alone (4.88)

₹228 cr, same seeds, 95% CI ± 0.06

All results: 30 replications × 30 days after 3-day warm-up, common random numbers, paired 95% confidence intervals.

PROJECT OVERVIEW

What we did, from proposal to recommendations

1

Proposal

Problem, questions, 7 what-if scenarios, KPIs (submitted deck)

2

Data collection

Timetable of 115 trains, rules, audit reports, CEA coal data

3

Input modelling

100 parameters; distributions where no records exist

4

Model build

AnyLogic 8.9.9, built in 3 phases: core, crew, brake certificate

5

Verify & validate

Invariants every event; benchmarks for cycle and throughput

6

Experiments

~60 cases × 30 replications; 2×2×2 factorial; sensitivity

Deliverables

AnyLogic model: RakeFlow.alp with 4 experiments, 100 parameters

Support files: parameter workbook, 15 research notes, 8 data templates

Analysis: test harness, experiment scripts, result tables and figures

This presentation and the submitted proposal deck

CONTEXT

A power station short of coal, fed by a 14 km single line

Closed circuit: 8 rakes of 58 BOXN wagons (~3,990 t) shuttle between WCL sidings at Ghugus and CSTPS Chandrapur.

Route: 14.11 km single line Ghugus–Tadali, then 13 km on the busy Wardha–Balharshah main line.

Constraints: crew duty 9 h, brake certificate 6,000 km / 30 days, demurrage after 3 h loading / 7 h unloading.

Investment on the table: Railways sanctioned ₹228 crore in July 2026 to double the single line.

Our question: running time is under an hour, the round trip is ~40 h. Where does the time go, and what raises coal delivered per day?

6–17 days

coal stock at CSTPS

norm 20–26 days (CEA)

6–10

rakes/day received

2,920 MW plant, rail-fed

145–163%

main-line utilisation

Wardha–Balharshah, 2023

₹228 cr

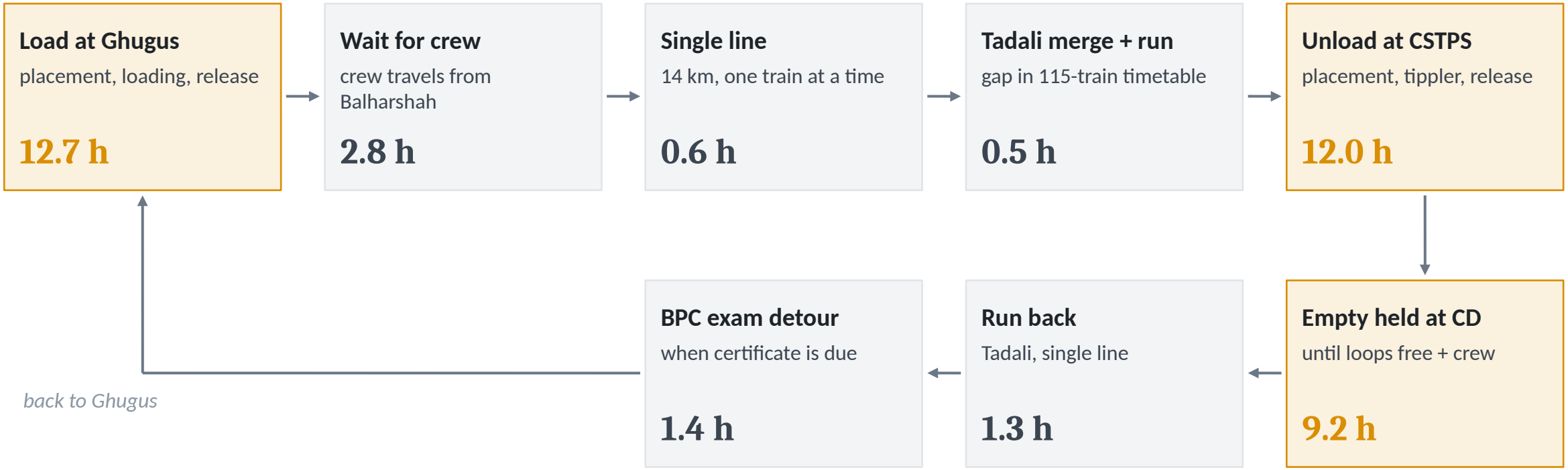
doubling sanctioned

+4.5 MTPA planned

Sources: CEA daily coal stock reports; Gazette of India 4 Jun 2026; RailPost 17 Jul 2026; The Hitavada 2023.

THE SYSTEM

One rake's round trip: eight steps, 40.3 hours



Three clocks run on every rake: crew duty (9 h, 11 h with notice, then relief) · brake power certificate (6,000 km / 30 days, 8 h daylight exam) · demurrage (₹150 per wagon-hour after free time)

Hours: baseline mean per cycle from the model (30 replications). Highlighted steps are the three largest.

What we tried to improve, and how we measured it

Objective maximise rakes/day to CSTPS and minimise mean cycle time, within crew-duty and brake-certificate rules

KPI	Type	Better	Baseline	Definition in the model
Rakes/day to CSTPS	Primary	↑	4.77	loaded rakes placed at CSTPS per day after warm-up
Coal delivered, t/day	Primary	↑	19,000	rakes/day × 3,990 t
Mean and P90 cycle time	Primary	↓	40.3 / 47.1 h	Ghugus arrival to next Ghugus arrival, per rake
Cycle-time breakdown	Diagnostic	—	20 stages	hours per cycle in each stage (load, crew, line, merge …)
Demurrage per rake	Cost	↓	₹33,300	₹150/wagon/h over free time, graded slabs (DM-01)
CSTPS stock, lowest	Outcome	↑	5.0 days	daily stock: receipts + non-rail supply − burn
Utilisation, queues	Diagnostic	—	line 21%	single line, sidings, tipplers; Tadali queue length
Crew expiries, exam hours	Constraint	≈ 0	0.02 / 100 trips	duty expiries per 100 trips; exam h/rake/month

MODEL FEATURES

What the AnyLogic model represents

Real timetable
115 passenger trains on their days of run, lognormal lateness, Poisson goods trains

Single-line block
one train at a time; enters only if a loop is free at the far end

Loops and sidings
2 Ghugus loops, 2 Tadali loops, 2 loading sidings; deadlock-free rules

Crew duty clock
sign-on, travel, 9 h / 11 h duty, relief, 12–16 h rest, pool of 16

Brake certificate
6,000 km / 30 days; depot detour; 8 h exam in daylight only

Demurrage
free time from placement; graded ₹150/wagon/h slabs at both ends

Plant side
4 of 5 tipplers; CSTPS coal stock with burn and non-rail supply

Measurement
20-stage cycle breakdown; per-rake cycle log in the data-template format

Built in three phases (1 core cycle, 2 + crew, 3 + brake certificate); each phase verified before the next.

PARAMETERS

Key parameters and where each value comes from

Parameter	Value	Type	ID
Fleet in the closed circuit	8 rakes	assumption	A-18
Rake: wagons / payload	58 / 3,990 t	hard	CR 2024-25
Ghugus loops / Tadali loops	2 / 2	assumption	A-01, A-02
Loading sidings at Ghugus	2	inferred	LD-02
Single line length	14.11 km	hard	INF-01
Tadali–Chandrapur distance	13 km	soft	INF-08
Tipplers in service	4 of 5	assumption	A-17, UL-01

Parameter	Value	Type	ID
Crew duty / with notice	9 h / 11 h	hard	CR-01, 02
Crew rest at HQ	12 h / 16 h	hard	CR-04
Crew pool, booking lobby	16, Balharshah	assumption	A-25, A-06
BPC validity	6,000 km / 30 d	hard	BPC-01
Free time load / unload	3 h / 7 h	hard	FT-01, 02
Demurrage rate	₹150/wagon/h	hard	DM-01
Warm-up / run / reps	3 d / 30 d / 30	design	brief 4.1

100 parameters in total, each on the AnyLogic canvas and named by its source ID: hard = official rule or record, soft = secondary source, assumption = no source, tested in sensitivity runs.

DATA SOURCES

All inputs come from public, cited sources

Source	What we extracted	How the model uses it
erail.in timetable	115 passenger trains: times at Tadali and Chandrapur, days of run	exact blocking schedule for the Tadali merge
etrain.info running history	lateness samples, 26–86 min	lognormal lateness for every train
Railway Board letters (RBE 143/2016 ...)	crew duty, rest, free time, demurrage slabs	hard rules in the logic
IRIMEE / JPO freight examination	BPC 6,000 km / 30 days, 8 h exam, daylight	exam trigger, duration, time window
CAG Reports 31/2014, 47/2015	yard detention times, crew sign-on delays	triangular min / mode / max
CEA daily coal stock reports	CSTPS consumption, receipts, stock	coal-stock KPI and validation target
Gazette, CR highlights, RailPost	14.11 km line, 58 wagons, 3,990 t, ₹228 cr	fixed inputs and the doubling scenario

Data gap: no Central Railway / WCL rake records were received. Eight collection templates and the model's cycle-log export are ready for them.

INPUT DISTRIBUTIONS

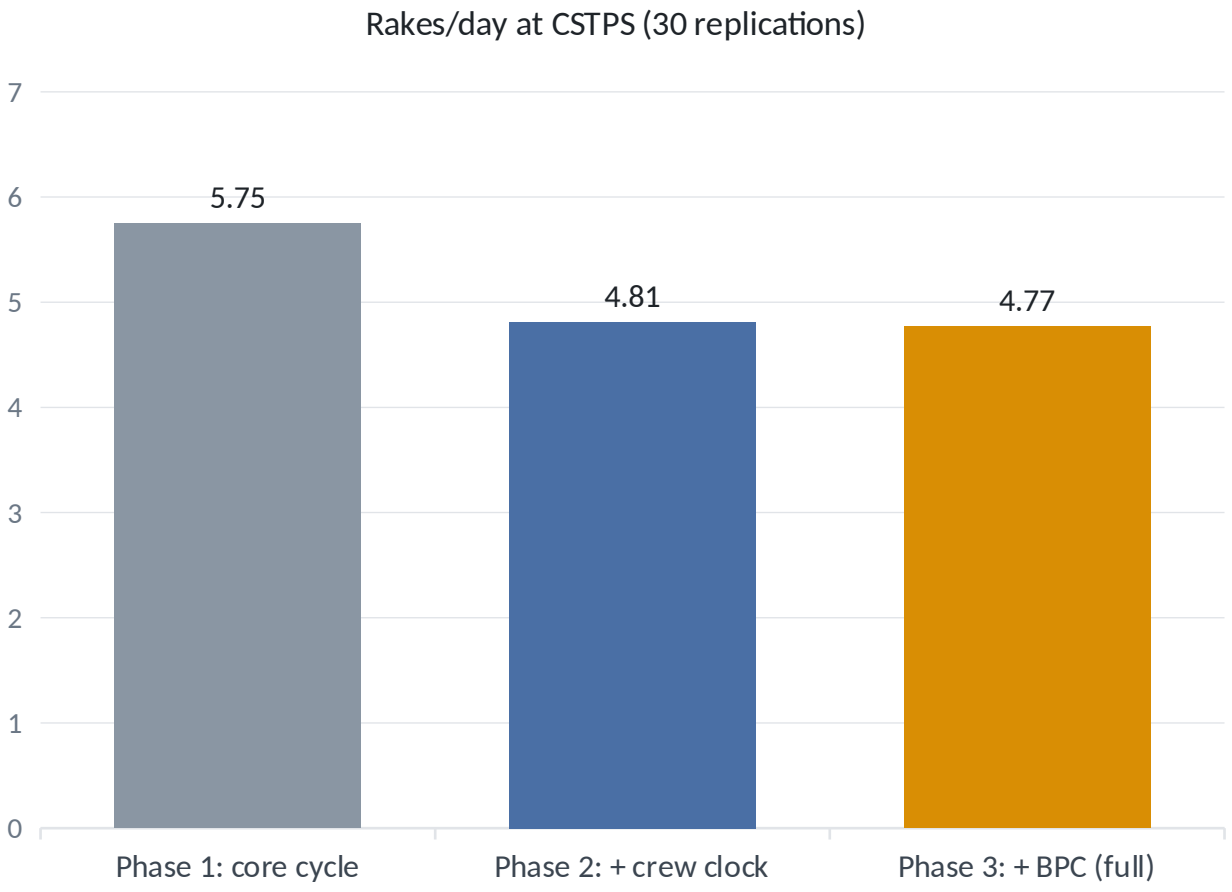
Where no record exists: triangular from min / most likely / max

Activity	Distribution	Basis
Loading, pay-loader / CHP	Tri(4, 6, 9) h	RB free time, manual norm
Loading, rapid silo (RLS)	Tri(1, 1.5, 3) h	PIB, MCL silos
Arrival → placement, Ghugus	Tri(0.5, 1, 4) h	CAG Table 18
Release → ready, Ghugus	Tri(0.8, 1.5, 6) h	CAG Table 18
Single-line block occupancy	Tri(25, 31, 38) min	speed + line clear
Main-line freight speed	Tri(30, 40, 50) km/h	IR goods average
Gap to a train at Tadali	Tri(6, 10, 15) min	block headway
Passenger lateness	Lognormal(30, 30) min	etrain samples

Activity	Distribution	Basis
Arrival → placement, CSTPS	Tri(0.6, 2, 8) h	CAG Table 18
Unloading by tippler	Tri(2, 4, 7) h	industry, RB 7 h
Release → despatch, CSTPS	Tri(1.5, 3, 8) h	CAG Table 18
Crew sign-on	Tri(0.5, 0.75, 2) h	CAG 47/2015
Crew travel to Ghugus	Tri(1, 1.5, 2.5) h	~40 km by road
Relief crew arrival	Tri(1, 2.5, 6) h	assumed
BPC exam duration	Tri(6, 8, 12) h	JPO 8 h + repairs
Through goods trains	Poisson, 20/day/dir	Ballarshah ~50/day

Triangular as proposed in the submitted deck; lognormal because lateness is positive and right-skewed; replace each with a fitted distribution when field data arrives.

Built in three phases; each added clock costs throughput



What each phase adds

Phase 1, core: loops, sidings, single line, Tadali paths, tipplers. 5.75 rakes/day, 33.4 h cycle.

Phase 2, crew: crews travel from Balharshah under the 9 h rule. Throughput falls to 4.81; cycle 39.9 h.

Phase 3, BPC: exam detours in daylight only. 4.77 rakes/day, 40.3 h. This is the baseline.

The crew clock alone removes almost a rake a day: a first hint that crew timing is a lever.

The model is consistent and matches public benchmarks

Verification

Invariants checked after every event: loop, siding and tippler capacity; single line used one way at a time; crew never past 11 h; no rake lost.

Identical results in AnyLogic and the test harness for the same seed (seed 1: 4.70 rakes/day, 40.5 h).

Little's law holds: 8 rakes ÷ 40.3 h × 24 = 4.77 rakes/day.

Stress-tested over 15 extreme configurations with no violations.

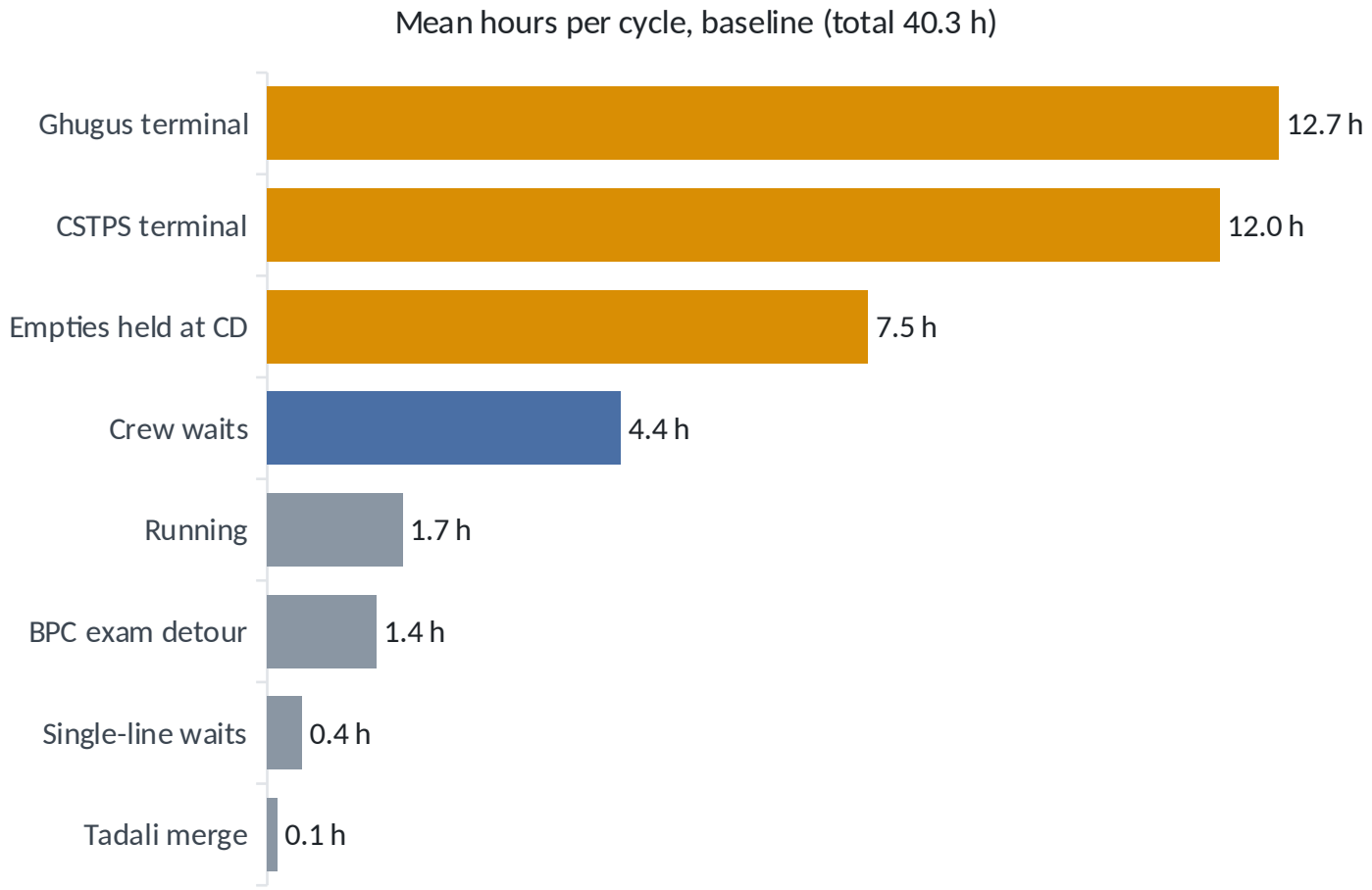
Validation

Measure	Benchmark	Model	Verdict
Mean cycle time	24–72 h	40.3 h	inside
Ghugus terminal detention	≤ 15–16 h (RB)	15.5 h	inside
Rakes/day, today's layout	6–10 (CEA)	4.77	below
Rakes/day, with 3 changes	7–8 (Ghugus)	7.76	inside
Crew duty expiries	rare	0.02 / 100	inside
Rakes over loading free time	many, not all	100%	all

Gaps trace to assumed Ghugus loops and yard times, and to loading time (min 4 h) exceeding the 3 h free time.

BASELINE RESULT

80% of the cycle is spent at the two terminals



32.2 h

at the terminals (80%)

Ghugus, CSTPS and empties held at CD

4.4 h

waiting for crews

travel from the Balharshah lobby

0.6 h

single line + merge

line 21% busy; Tadali queue 0.03

EXPERIMENT PLAN

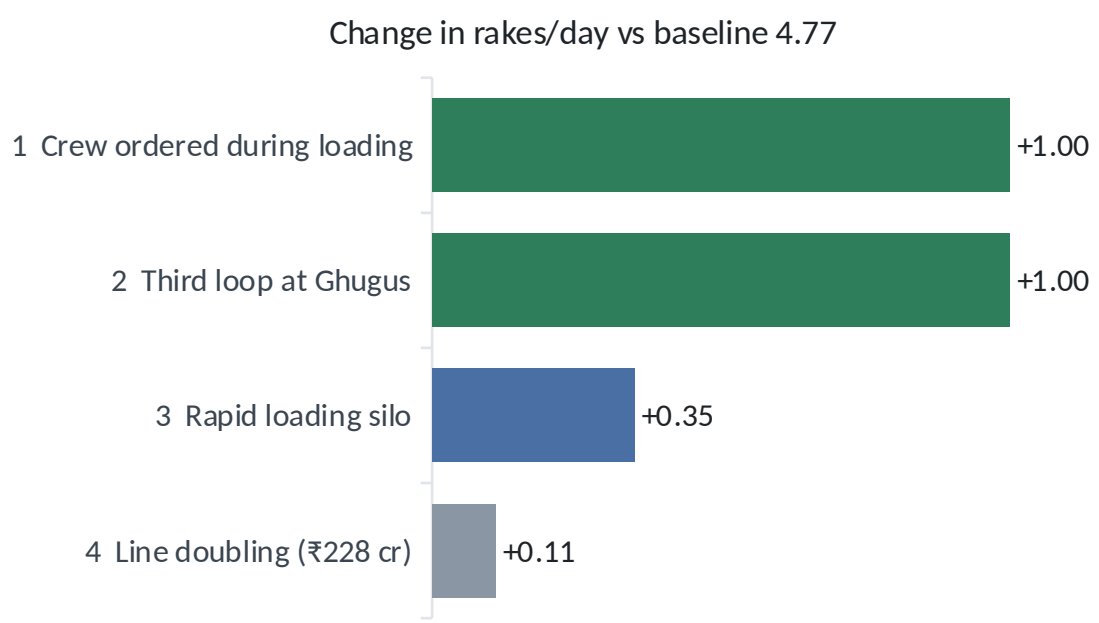
Every experiment we ran

Group	Factor varied	Levels	Cases
Key experiments	crew timing · Ghugus loops · loading method · line doubling	one change each vs baseline	5
2×2×2 factorial	crew timing × Ghugus loops × loading method	all 8 combinations	8
Proposal what-ifs	single-line dispatch · Tadali slots · exam trigger	alternate, batch; slot; 0/3/5/7 days	8
Fleet size (E1)	rakes in the circuit	4, 6, 8 … 20	9
Resource levers (E2–E7)	RLS, Chandrapur lobby, doubling, tipplers, loops, crew pool, sidings	+20% or one unit each	12
Assumption sensitivity	loops, yard times, CSTPS times, goods trains, lateness, crew pool, controller rule	one at a time	11
Model phases	core → + crew → + BPC	3 phases	3

Every case: 30 replications · 30 days after a 3-day warm-up · same 30 seeds as the baseline (common random numbers) · paired 95% t-interval

KEY EXPERIMENTS

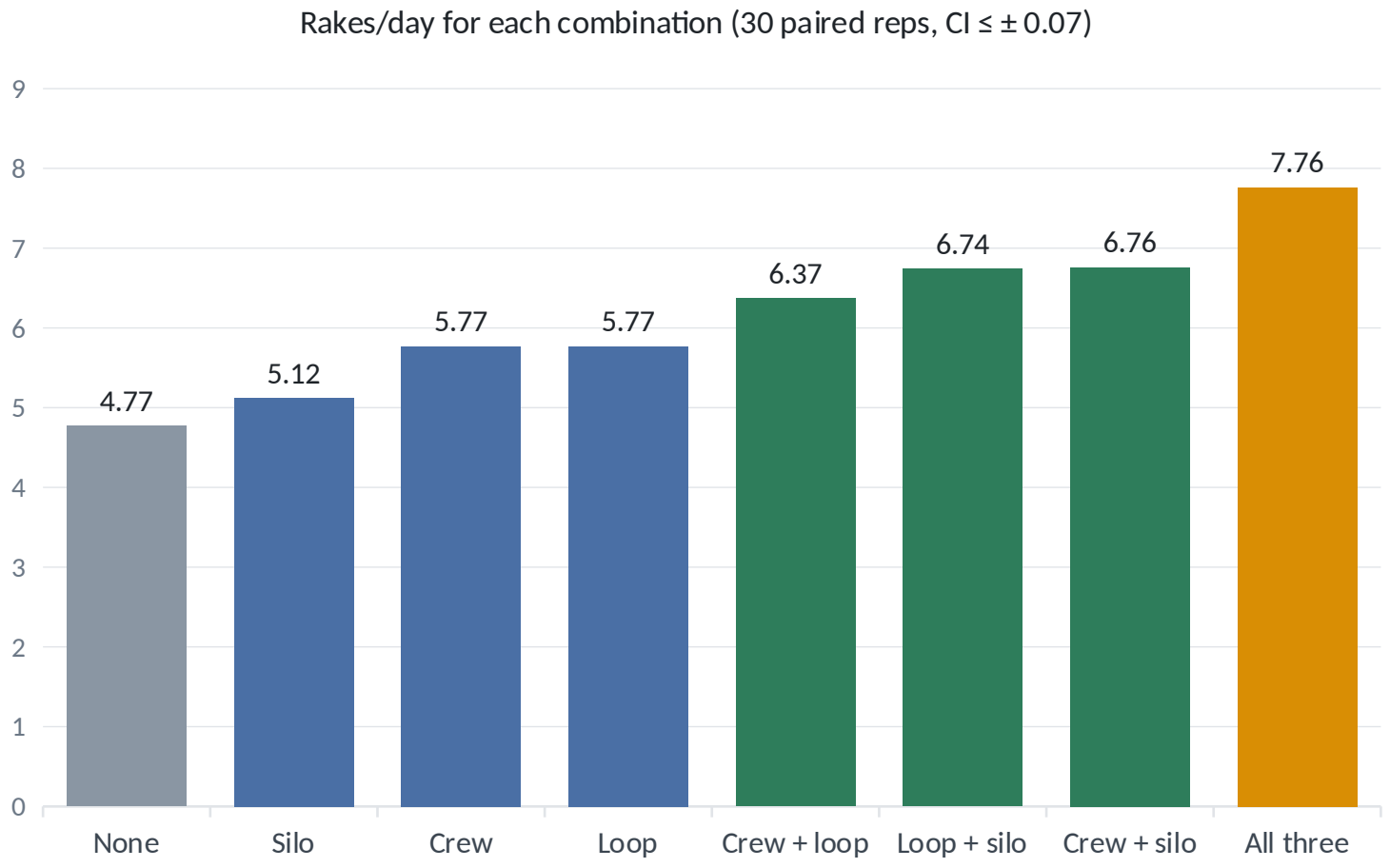
Crew timing and a third loop each add a full rake per day



Experiment	Rakes/ day	Δ (95% CI)	Cycle h	Demurrage
Baseline	4.77	—	40.3	₹33,300
1 Crew during loading	**5.77**	**+1.00 ± 0.06**	**33.3**	₹33,400
2 Third Ghugus loop	**5.77**	**+1.00 ± 0.09**	**33.3**	₹33,200
3 Rapid loading silo	5.12	+0.35 ± 0.07	37.5	**₹0**
4 Line doubling	4.88	+0.11 ± 0.06	39.4	₹33,300

Experiment	What changed in the model	What moved (hours per cycle)
1 Crew timing	crew booked when loading starts so it arrives as the rake is ready	crew wait 4.4 → 1.8; empties held at CD 7.5 → 3.8
2 Ghugus loop	Ghugus loops 2 → 3	empties held at CD 7.5 → 1.7
3 Loading silo	loading Tri(4, 6, 9) h → Tri(1, 1.5, 3) h	Ghugus terminal 12.7 → 6.4; but held at CD 7.5 → 10.6
4 Doubling	one line per direction Ghugus–Tadali	line use 21% → 11%; cycle −0.9

Combined, the three changes reach 7.76 rakes/day



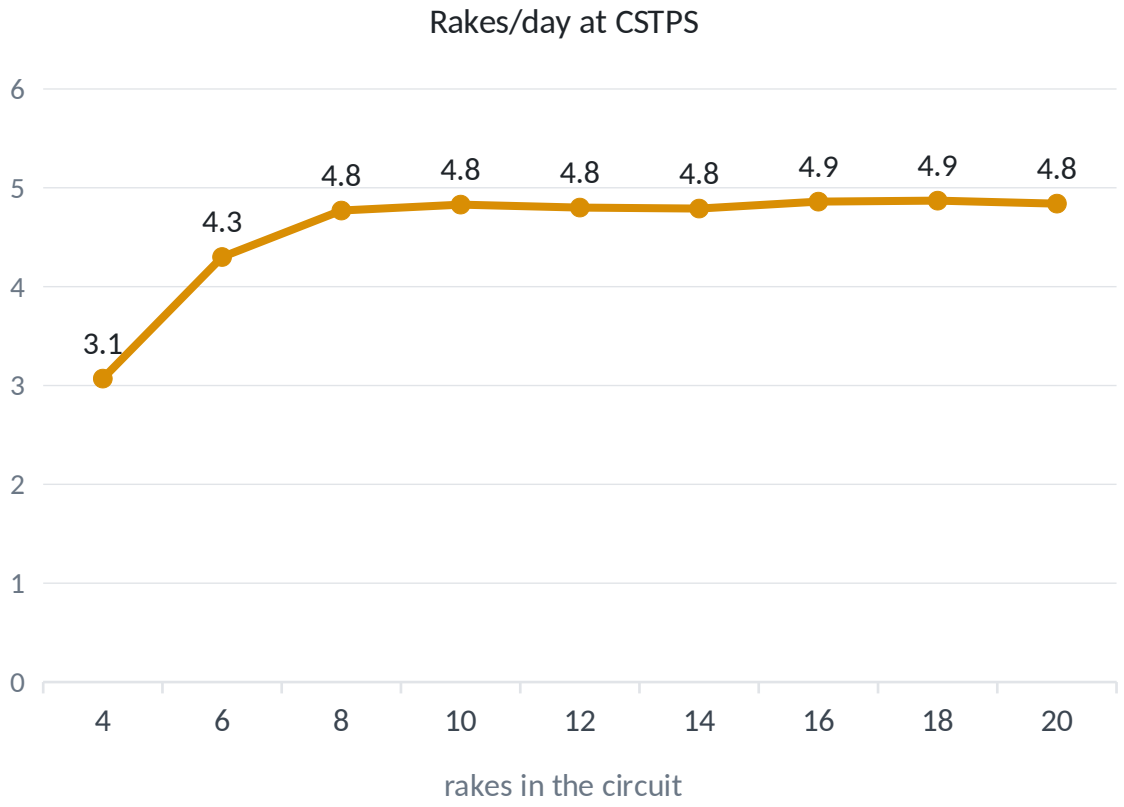
Estimated effects on rakes/day

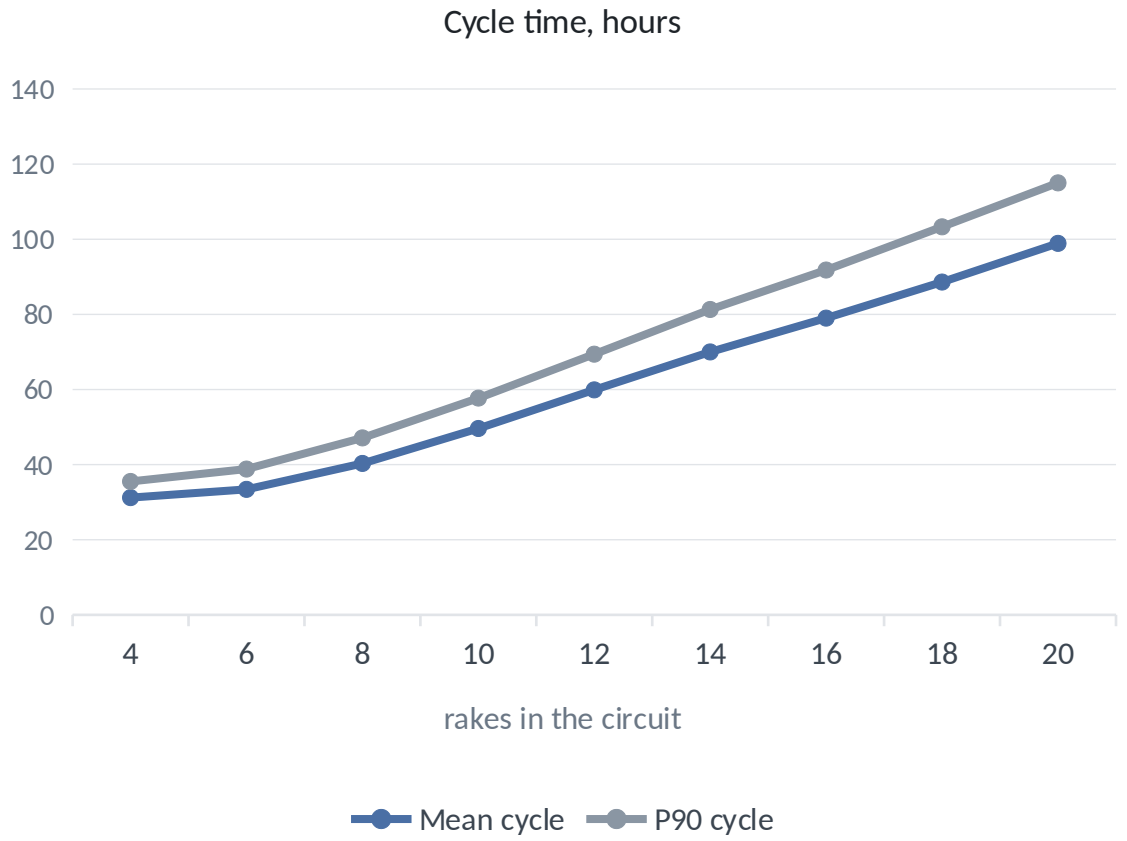
Effect	Δ rakes/day
Crew during loading	**+1.07**
Third Ghugus loop	**+1.06**
Rapid loading silo	**+0.92**
Crew × loop	**−0.26**
Crew × silo	**+0.27**
Loop × silo	**+0.25**
Crew × loop × silo	−0.06

Crew and loop overlap because both free loop space; the silo pays off once loop space is fixed.

FLEET SIZE (E1)

More rakes only lengthen the cycle





Throughput saturates at 8 rakes because the Ghugus loops are full; by Little's law extra rakes add only waiting time.

PROPOSAL WHAT-IFS

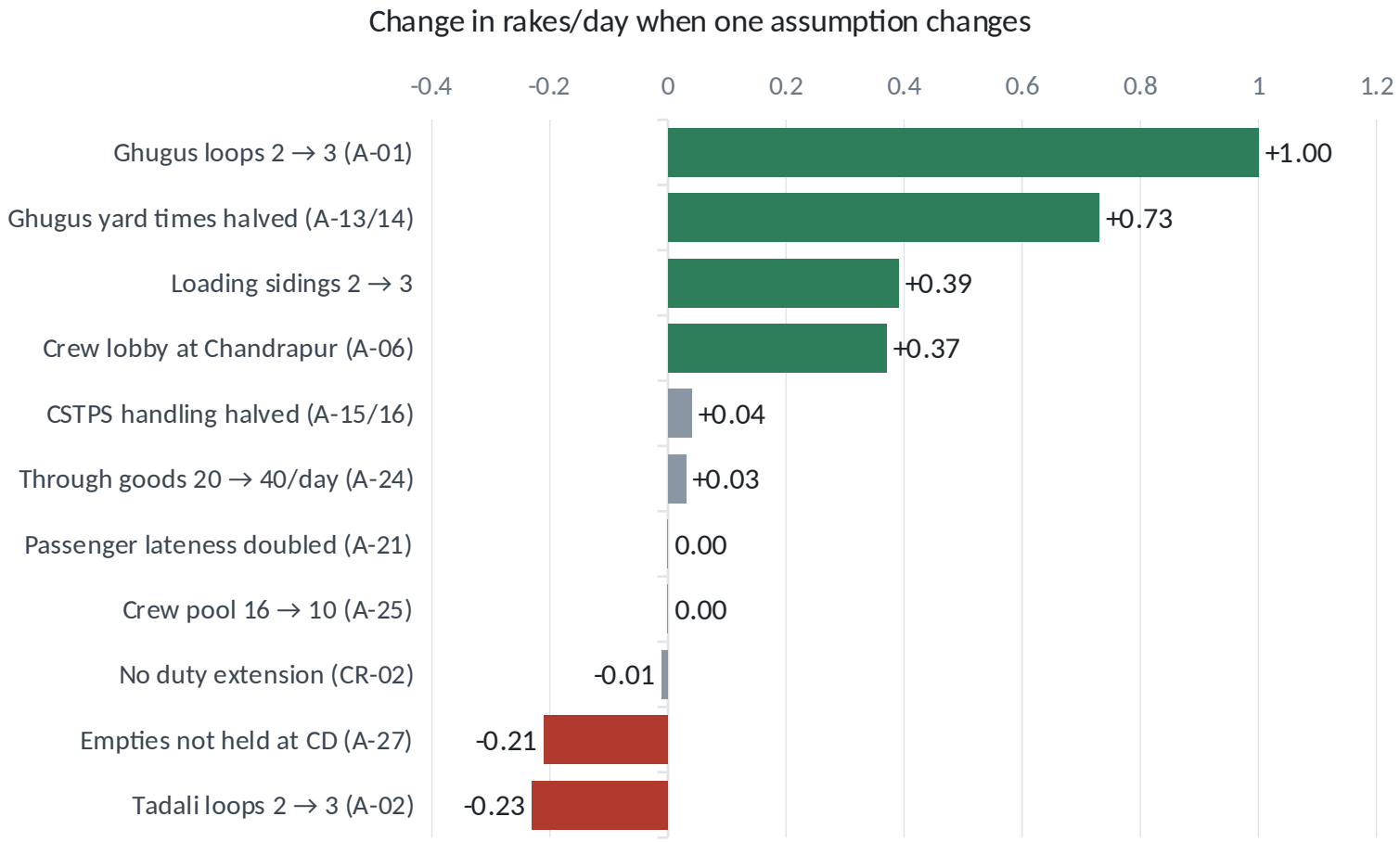
Results for the seven scenarios we committed to

#	Scenario from the proposal	Rakes/day	Δ vs base	Cycle mean / P90	Finding
1	Base case: loaded first, crew when ready	4.77	—	40.3 / 47.1 h	reference
2	Dispatch: alternating / batches of 3	4.77	0.00	40.3 / 47.1 h	no effect: line 21% busy
3	Tadali: reserved path vs take gaps	4.78	+0.01	40.2 / 47.1 h	no effect: merge wait 0.13 h
4	Crew booked during loading	**5.77**	**+1.00**	**33.3 / 38.3 h**	**largest free gain**
5	Exam trigger at expiry / 3 / 5 / 7 days	4.80–4.75	± 0.03	≈ 40 / 47 h	exam h/rake/month 23 → 30
6	Fleet size 8 → 20 rakes	4.77–4.87	≤ +0.10	40 → 99 h	throughput flat
7	Line doubling vs best policy	4.88 / 5.78	+0.11 / +1.01	39.4 / 33.2 h	policy beats track

Scenario 7: 'best policy' = crew booked during loading; with doubling added it reaches 5.78 rakes/day (+1.01), i.e. doubling adds nothing on top.

SENSITIVITY

Which assumptions the answer depends on



Reading this chart

Ghugus layout and yard times move the answer most: collect these first.

Main-line traffic and lateness barely matter: the merge is not binding.

A third Tadali loop hurts: it lets more empties crowd the Ghugus loops.

SCORECARD

Today vs the recommended package vs doubling

KPI	Today	Crew + loop + silo	Change	Doubling only
Rakes/day to CSTPS	4.77	**7.76**	**+63%**	4.88 (+2%)
Coal delivered, t/day	19,000	**31,000**	**+12,000 t**	19,500
Mean rake cycle	40.3 h	**24.7 h**	**−39%**	39.4 h
P90 rake cycle	47.1 h	**28.6 h**	**−39%**	45.9 h
Demurrage per rake	₹33,300	**₹0**	**−100%**	₹33,300
CSTPS stock, lowest	5.0 days	**10.4 days**	**× 2**	5.4 days
Single-line utilisation	21%	34%	still low	11%
Crew expiries / 100 trips	0.02	**0**	rule kept	0.01

30 paired replications per column; 95% CI on rakes/day ≤ ± 0.06.

RECOMMENDATIONS

What to do, in order of cost

1. **Book crews when loading starts**
   free · +1.0 rake/day · −7 h cycle

2. **Add a third loop at Ghugus**
   small yard work · +1.0 rake/day (with 1: +1.6)

3. **Install a rapid loading silo**
   removes ₹33,300 demurrage per rake · all three: +3.0 rakes/day

4. **Treat the doubling as a growth project**
   +0.11 rake/day for today's traffic

5. **Keep the fleet at 8; examine rakes near expiry**
   more rakes or early exams only add waiting

Limitations and next steps

Assumed layout: Ghugus loops and yard times drive the result; confirm with Central Railway.

Dispatcher rule: empties held at Chandrapur (A-27) is our assumption.

Closed circuit only: other traffic that would use a doubled line is not modelled.

Next: fit distributions to CR / WCL rake logs using the ready templates; rerun.

Fix the terminals before the track

Three changes at Ghugus deliver 63% more coal per day and a 39% shorter cycle; the ₹228 crore doubling adds 2%.

+1.00
crew booked during loading

+1.00
third Ghugus loop

7.76
rakes/day with all three

Thank you · Questions